

Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | [linkedin.com/in/varunprasaath-rs](https://www.linkedin.com/in/varunprasaath-rs)

PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years building structure-property-processing datasets across organic, polymer, and thin-film material systems. Experienced in solid-state-adjacent characterization, thermal/spectroscopic analysis, stability/degradation studies, DOE/SPC, and technical reports that connect material attributes to process and formulation decisions.

CORE COMPETENCIES

Pharmaceutical Materials Science

Solid-State Characterization

Crystallization / Stability Mindset

Thermal Analysis

FT-IR / UV-Vis / NMR / MS

DOE/SPC

Technical Reports

Cross-Functional CMC Readiness

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided process-development experiments for bio-based polymer systems, defining material-quality readouts, process windows, reproducibility criteria, and SOP-ready specifications for scale-up.
- Use FT-IR, thermal analysis, spectroscopy, and statistical process-control concepts to connect process parameters with material stability, quality attributes, and method repeatability.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed, synthesized, purified, and characterized 15+ organic electrochromic material variants using UV-Vis-NIR, FT-IR, cyclic voltammetry, spectroelectrochemistry, NMR, GPC, TGA, DSC, and mass-spectrometry-informed analysis.
- Mapped structure-property-stability relationships, degradation pathways, and performance failure modes to guide molecular design and analytical interpretation.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded molecular thin-film process development, linking CVD/surface preparation to morphology, interfaces, optical performance, and reproducible process controls.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized polymer thin-film deposition using DOE strategies and in-situ QCM-D process monitoring, characterizing film-growth kinetics and coating reproducibility.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented equipment tracking, calibration routines, SOPs, and safety protocols that improved instrument uptime by 30% and strengthened data integrity.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Pharmaceutical Materials Characterization Bridge Target Role

Translate organic/materials chemistry evidence into API/excipient, polymorph, crystallization, amorphous-solid-dispersion, and physical-stability problem solving.

FT-IR, UV-Vis-NIR, NMR, GPC/SEC, MS-informed analysis, TGA/DSC, DOE

Analytical Stability and Degradation Study Chemistry of Materials 2025

Identified degradation, redox instability, and performance failure modes across functional material variants using orthogonal analytical characterization.

UV-Vis-NIR, FT-IR, CV, TGA/DSC, NMR, GPC, MS-informed analysis

DOE-Guided Polymer Process Development Current

Defined process windows, analytical readouts, reproducibility criteria, and SOP-ready specifications for scalable polymer systems.

DOE, SPC, FT-IR, thermal analysis, technical reports

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Materials & Solid-State Characterization: FT-IR, UV-Vis-NIR, ellipsometry, NMR, GPC/SEC, GC-MS/LC-MS exposure, QCM-D, TGA, DSC, cyclic voltammetry; XRD/SEM-adjacent familiarity

Pharmaceutical Materials Framing: Crystallization/polymorph/salt-selection concepts, physical stability, drug-excipient compatibility mindset, amorphous-solid-dispersion awareness, structure-property-processing analysis

Process & Statistical Methods: DOE, SPC, JMP/basic, Excel, Python/basic, process windows, reproducibility studies, SOP-ready specifications

Documentation & Communication: Technical reports, scientific publications, data packages, cross-functional presentations, lab notebooks, regulatory-documentation readiness

JOURNAL PUBLICATIONS

- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.
- Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.
- Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)